Ryan Eng

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EDUCATION

Brown University, *Sc.B., Computer Science-Applied Math*; *Sc.M., Computer Science, AI/Machine Learning Track* **Expected 2027**

Coursework: Deep Learning (Graduate), Prescriptive Analytics/Optimization (Graduate), Quant Finance I (co-led) **GPA:** 3.9/4.0
Algorithmic ML, Automated Decision Making, Information Theory, Computer Networks, Systems for ML, Operations Research, Database Systems, Software Engineering, Comp. Linguistics, Data Science, Crisis Econ, Computer Systems

Activities: Quantitative Trading at Brown, Full Stack @ Brown (*PM*), Presidential Host (*Technology Consultant*), IMC Prosperity (*Top 3% of 9K teams*), Data Science Institute (*TA, Board Member),* Brown Daily Herald (*Business*)

QUANTITATIVE EXPERIENCE

Amazon Web Services Inc., *Cloud Engineer Intern* **Jun. 2026 – Aug. 2026**

- Designed high-throughput metric and histogram recorders to analyze time series and distributional latency data on system ops
- Engineered diagnostic pipelines to decouple client-side and server-side execution failures and forecast performance degradation
- Formulated tail degradation and rolling short term and long term latency algorithms (quadratic mean) to highlight bottlenecks during peak seasonal demand, transient errors, and potential optimizations for underutilized Kubernetes and caching schema
- Architected E2E telemetry aggregation by designing relaxed, lock free atomic counters and a batch flushing protocol, processing and parallelizing high-throughput hot path ops with ~0 CPU memory overhead (*mean ~15ms, std 3.3ms*)

Trexquant Investment LP, *Quantitative Researcher* **Sep. 2025 – Oct. 2025**

- Feature engineer and apply machine learning methods to generate signals for Trexquant Earnings Call strategies (Fall 2025)

Stealth Startup, *Co-Founder* **Apr. 2025 – Sep. 2025**

- Optimize company verticals via Contextual Retrieval (*RAG*) large language model (*LLM*) architecture (*OpenAI, Langchain*)
- Create data lake, shard, and graph network on *AWS cloud* + *bedrock* to catalogue data for LLM retrieval, and calibration
- Finished in top 10% of startups for Y-Combinator Venture Capital summer cohort

JL Sports (National Football League), *Quantitative Analyst* **Mar. 2025 – Present**

- Guided NFL-affiliated agency in acquisition of 2026 draft-eligible prospects through quantitative scouting research
- Implemented logistic regression + random forest models to mirror NFL scouting methodologies (90% accuracy on past drafts)

Brown University Department of Computer Science, *Reinforcement Learning Research Assistant* **Jun. 2024 – Sep. 2024**

Brown University School of Public Health, *Software Developer, Research Assistant* **May 2024 – Mar. 2025**

LEADERSHIP EXPERIENCE

Introduction to Quantitative Finance I, *Instructor and Course Developer* **Apr. 2024 – Dec. 2024**

- Designed and co-instructed Brown Fall 2024 Independent Study Course under Professor Spencer Yongwook Kwon
- Curated content on regression (*AR, Newey-West*), market microstructure, stat-arb, factor modeling (*Fama French*), PCA

Quantitative Trading at Brown (QTAB), *President, Researcher, Software Lead (Backtester and Trading Game)* **Sep. 2023 – Present**

TECHNICAL SKILLS

- **Proficient:** Python, Java | **Intermediate:** Go, C, C++, Typescript, JavaScript, HTML, CSS, SQL | **Basic:** YAML, Bash
- **Tools:** Data Science: Scikit-learn, Pandas, BS4, Selenium | Deep Learning: TensorFlow(Keras), PyTorch | Database: Redis, TimescaleDB, InfluxDB, Apache Spark, Docker, Kubernetes, Firebase | Other: AWS, Git, React, Flask, Next.js, Socket.io

SELECT PROJECTS AND INTERESTS

- **RI Flood Detection Optimization (Honors Thesis)** – inform sensor placement for climate response by probabilistically modeling regional flood risk in mutual information optimization objective. Advised by Professor Emanuele Di Lorenzo
- **Dynamic Factor Model Risk Optimizer** – modeled systematic risk (*Barra*) and sector-based factor loadings of equities from 4.2M datapoints for QTAB and Brown Healthcare Investment Group's portfolio; stored on TimescaleDB for querying
- **Sample Efficient Multi-Modal Reinforcement Learning** – a self-predictive reinforcement learning model that achieved 1/2 the human performance benchmark from ~25 minutes (1/8 the benchmark training time) of embedded audio/pixel input training
- **Replicating Value Factor in Cryptocurrency** – modeled "value" factor by feature engineering 47 signals (autoregressive and characteristics) in replication of Stanford research; used Metcalfe's law, hash-rate, transaction fees as "value" proxies

Interests: Soccer (English Premier League), Sports Analytics (Director of Analytics at Brown Sports Network), Track, Swimming, Music, Volunteering, Physics (Microsoft Quantum Computing)